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ORIGINAL PAPER

Land degradation dynamics and resilience of tropical montane peatlands in the Serra do Espinhaço Meridional revealed by the SDG 15.3.1 indicator

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Abstract Tropical montane peatlands are critical ecosystems for carbon storage, hydrological regulation, and biodiversity conservation, yet they remain poorly studied in tropical regions. In this study, we assessed land degradation dynamics in peatlands of the Serra do Espinhaço Meridional, southeastern Brazil, using the Sustainable Development Goal (SDG) Indicator 15.3.1 integrated with remote sensing data, spatial statistics, and environmental modeling. The indicator combines three sub-components primary productivity, land cover, and soil organic carbon derived from MODIS, ESA CCI, and SoilGrids datasets for the period 2001–2020. Spatial autocorrelation was evaluated using global and Local Moran's I, and the drivers of degradation, stability, and improvement were analyzed through multinomial logistic regression incorporating a spatial lag term. The results indicate that most peatlands in the study area remain stable or show signs of improvement, while degradation is spatially clustered and restricted to specific areas. Hydrological and topographic variables emerged as the main factors controlling peatland condition, highlighting the spatially structured nature of degradation and recovery processes. Our findings demonstrate the resilience of tropical montane peatlands under current land-use conditions and confirm their role as long-term carbon reservoirs. The SDG 15.3.1 indicator proved to be an effective tool for diagnosing peatland condition and identifying priority areas for conservation. This study provides a scalable and spatially explicit framework to support peatland monitoring, conservation planning, and the implementation of sustainable land management policies in tropical mountain regions.

Keywords Tropical peatlands · Land degradation · SDG 15.3.1 · Spatial autocorrelation · Soil organic carbon · Remote sensing

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Introduction

Peatlands are wetland ecosystems characterized by the accumulation of partially decomposed organic material, playing a fundamental role in biodiversity conservation and the provision of ecosystem services (Silva et al. 2022). These environments exhibit high water-retention capacity and act as important carbon sinks, contributing significantly to climate regulation

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and water quality maintenance (Rocha Campos et al. 2016). In the Serra do Espinhaço Meridional (SdEM), located in Minas Gerais, Brazil, peatlands are essential components of the landscape and support unique biological diversity, which led the region to be recognized by UNESCO as a Terrestrial Biosphere Reserve (Silva 2013; Kuchenbecker 2019).

Despite their ecological importance, peatlands in the SER are increasingly threatened by climate change, land-use conversion, and the unsustainable exploitation of natural resources. The intensification of deforestation and agricultural activities has promoted ecosystem degradation, compromising hydrological functions, carbon storage capacity, and biodiversity conservation (Silva et al. 2022). Importantly, these degradation processes are spatially heterogeneous across the SER, highlighting the need to identify areas under greater anthropogenic pressure to support effective monitoring, conservation, and management strategies (Rocha Campos et al. 2016; Silva et al. 2022).

In this context, environmental indicators represent an essential approach for characterizing ecosystem condition and monitoring degradation processes. Several indicators have been proposed to assess ecosystem health and environmental change (Karr 1981; Mazzoni 2013), among which the Sustainable Development Goal (SDG) Indicator 15.3.1 has gained prominence at the global scale (United Nations 2015, 2020). This indicator integrates three complementary sub-indicators—vegetation productivity, land cover change, and soil organic carbon—providing a comprehensive framework for identifying areas susceptible to land degradation (United Nations 2020; Conservation International 2023).

The SDG 15.3.1 Indicator has been applied across a wide range of ecosystems, including boreal forests, savannas, arid regions, and watersheds, to assess trends in land degradation, recovery, and sustainable land management (Von Maltitz et al. 2019; Easdale et al. 2019; Kust et al. 2020; Trinfonova et al. 2021; Hu et al. 2021; Jiang et al. 2022). In addition to supporting environmental monitoring, the indicator has been widely used to inform public policies aimed at mitigating land degradation, promoting ecological restoration, and advancing sustainable land-use practices (Easdale et al. 2019; Hu et al. 2021).

Despite its broad application, studies applying SDG 15.3.1 specifically to peatland ecosystems are still lacking, representing a relevant gap in the literature. Addressing this gap is particularly important given the high sensitivity of peatlands to environmental pressures and their strategic role in carbon sequestration and hydrological regulation. To our knowledge, this is the first spatially explicit application of the SDG 15.3.1 indicator to tropical montane peatlands. Therefore, this study provides a novel contribution by applying the SDG 15.3.1 Indicator to assess land degradation in the peatlands of the Serra do Espinhaço Meridional. The aim is to characterize the conservation status of these ecosystems and to support the development of conservation, restoration, and sustainable management strategies, contributing to the maintenance of the essential ecosystem services provided by peatlands.

Methods

Study area characterization

The Serra do Espinhaço Meridional (SdEM) is located in the state of Minas Gerais, southeastern Brazil (Fig. 1), extending approximately 300 km in a north–south direction and encompassing the territory of 26 municipalities. The region is recognized for its environmental and hydrological relevance, as it constitutes a major watershed divide hosting the headwaters of three important river basins in eastern Brazil: the Jequitinhonha, São Francisco, and Doce basins. These headwaters are closely associated with tropical montane peatlands, ecosystems that play a key role in hydrological regulation and long-term carbon storage (Silva et al. 2013; Silva et al. 2022).

According to the Köppen climate classification, the SER presents a tropical highland climate (Cwb), characterized by humid summers, dry winters, and mild temperatures throughout the year (Alvares et al. 2013; Silva et al. 2022). These climatic conditions, combined with pronounced topographic heterogeneity, favor high environmental diversity and the coexistence of two major Brazilian biomes: the Atlantic Forest and the Cerrado. Vegetation formations include campos rupestres, moist and dry grasslands, cloud forests, and Cerrado physiognomies, contributing to the region's high biodiversity (Gonzaga and Machado 2021; Silva et al. 2022).

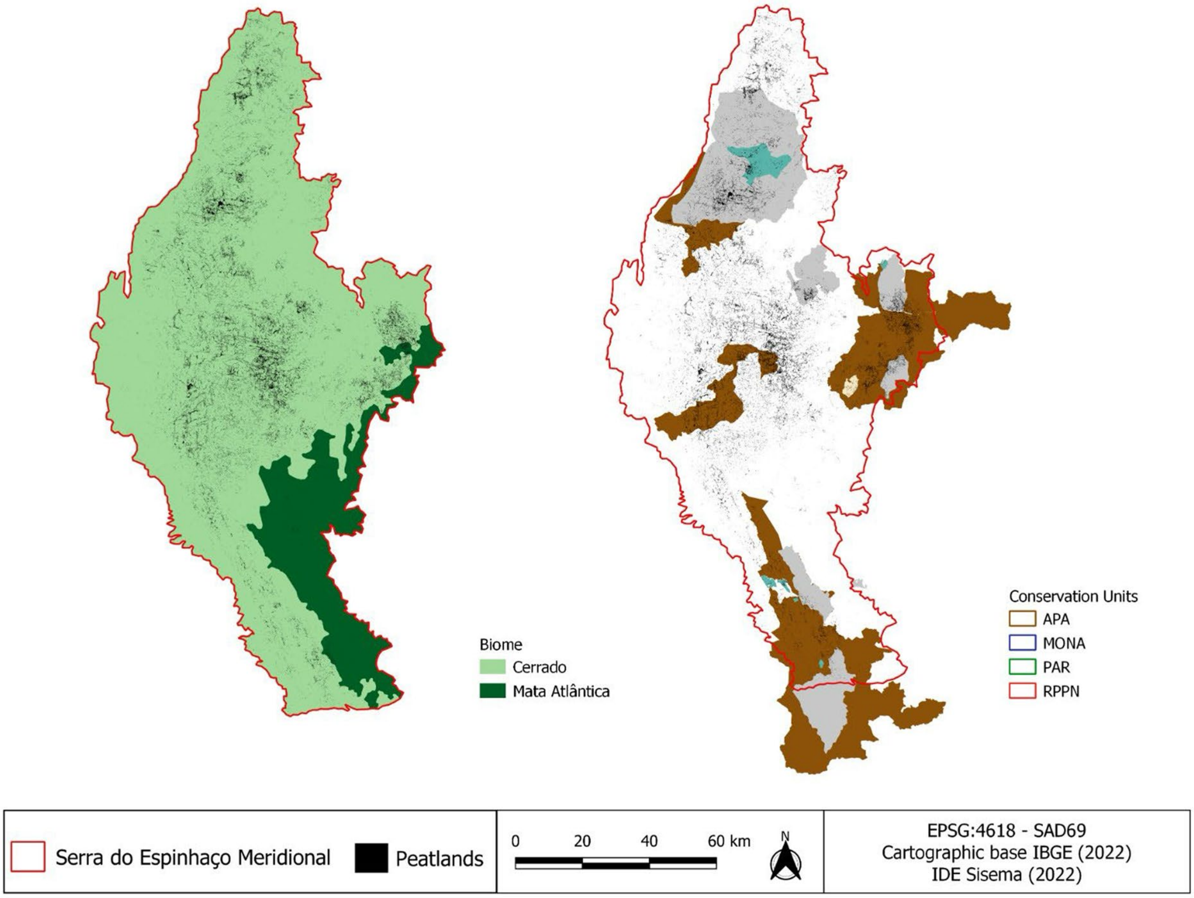

Fig. 1 Serra do Espinhaço Meridional in Minas Gerais, Brazil: **A** Biomes and **B** Protected Areas: APA—Environmental Protection Area; MONA—Natural Monument; PAR—Park; RPPN—Private Natural Heritage Reserve. Peatland extent adapted from Brandão-Martins (2023)

Peatlands in the SER are predominantly associated with Organosols, characterized by high contents of humified organic matter and prolonged soil water saturation. These soils play a fundamental role in water retention, microclimate regulation, and carbon cycling, acting as important carbon sinks with relevance to global climate regulation (Silva et al. 2009; Horák-Terra et al. 2014).

Assessment using the SDG (Sustainable Development Goal) 15.3.1

Peatland distribution in the SdEM was obtained from the mapping produced by Brandão-Martins (2023), based on Sentinel-1, Sentinel-2, and Landsat-8 imagery. Multiple spectral indices were applied to enhance the detection of tropical montane peatlands, using images from the dry season (May–October 2019), which improves the discrimination between permanently saturated peatlands and seasonally wet environments. Classification was performed using a Random Forest algorithm trained with visually interpreted samples and field data from previous studies. The mapped peatland area ranged between 21,000 and 26,000 ha across the SdEM.

The conservation status and land degradation of peatlands were assessed using the SDG 15.3.1 Land Degradation Indicator, linked to Sustainable Development Goal 15 (United Nations 2015). This indicator quantifies the proportion of degraded land relative to the total land area and integrates three sub-indicators: vegetation primary productivity, land cover change, and soil organic carbon (SOC) (Conservation International 2023). All analyses were performed using the Trends.Earth plugin within the QGIS environment (QGIS.org 2022).

Primary productivity

Land productivity refers to the biological capacity of land to sustain biomass production and constitutes the basis for food, fiber, and fuel generation essential for human survival (United Nations 2020). Primary productivity, in particular, is a key indicator for evaluating ecosystem dynamics, as it reflects the efficiency of vegetation in converting solar energy into biomass through photosynthesis. This sub-indicator was calculated based on the analysis of time series of the Normalized Difference Vegetation Index (NDVI) (Tucker 1979; Easdale et al. 2019; Giuliani et al. 2020).

NDVI was calculated using information from the red and near-infrared portions of the electromagnetic spectrum (Conservation International 2023), derived from MODIS sensor imagery with a spatial resolution of 250 m (NASA 2002). Biweekly NDVI data from 2001 to 2020 were used.

NDVI was calculated as:

$$NDVI = \frac{(NIR - RED)}{(NIR + RED)}$$

where *RED* corresponds to wavelengths between 620 and 670 nm and *NIR* to wavelengths between 841 and 876 nm.

Based on annual mean NDVI values, three complementary sub-indicators were derived: trajectory, state, and performance. The trajectory sub-indicator was obtained through linear regression of NDVI values over time, identifying positive, negative, or stable trends in vegetation productivity. The state sub-indicator compared current productivity levels with historical conditions to detect declines or stability. The performance sub-indicator compared local productivity to areas with similar land-cover characteristics using ESA CCI land-cover data (300 m resolution), allowing contextual evaluation of productivity within comparable environments (Ivits and Cherlet 2013; Hu et al. 2021; Reith 2021). The integration of these three components resulted in the final primary productivity classification.

Land cover

Land cover dynamics were evaluated using ESA CCI land-cover datasets (300 m resolution) for two reference years: 2001 (baseline) and 2020 (target year). Land-cover classes were reclassified into seven standardized categories adopted by the UNCCD: forest, grasslands, croplands, wetlands, artificial areas, bare land, and water bodies (Reith 2021).

Land-cover transitions between the two periods were evaluated using a transition matrix (Table 1), which classified changes as degradation, improvement, or stable conditions according to UNCCD guidelines (Conservation International 2023; Lambertin et al. 2023).

Soil organic carbon

Soil organic carbon (SOC) was assessed using the SoilGrids 250 m database, which provides spatially explicit estimates of carbon stocks for multiple soil depths based on machine learning models calibrated with global soil profile data and environmental covariates (Poggio et al. 2021; Turek et al. 2023). In this study, SOC stocks for the upper 0–30 cm soil layer were used, as recommended by the SDG 15.3.1 framework.

SOC variation between 2001 and 2020 was calculated following the methodology proposed by SDG Indicator 15.3.1 and implemented in the Trends.Earth platform (Conservation International 2023; Ghosh et al. 2024). Because direct long-term SOC monitoring is limited by high spatial variability, restricted temporal datasets, and the logistical complexity of repeated soil surveys, Trends.Earth applies a combined land cover and SOC approach to estimate changes in carbon stocks. This methodology integrates land cover transitions with baseline SOC data, using land cover as a proxy for land use, to estimate SOC changes over time (Conservation International 2023).

Land cover maps were first reclassified into the seven categories required by the United Nations Convention to Combat Desertification (UNCCD): forest, grassland, cropland, wetland, artificial areas, bare land, and water. Changes in land cover between baseline and target years were then linked to IPCC and UNCCD conversion coefficients that represent proportional SOC stock changes

Table 1 Land cover transition matrix indicating degradation (negative sign), improving (positive sign), or stable (zero) land condition (TRENDS.EARTH)

Land cover in initial year \ Land cover in target year	Forest	Grasslands	Croplands	Wetlands	Artificial	Bare lands	Water body
Forest	0	-	-	0	-	-	0
Grasslands	+	0	-	+	-	-	0
Croplands	+	+	0	+	-	-	+
Wetlands	-	-	-	0	-	-	0
Artificial	+	+	-	+	0	-	+
Bare lands	+	+	+	+	+	0	+
Water body	0	-	-	0	-	-	0

Legend

Degradation	Stable	Improving
-	0	+

associated with land-use transitions over a 20-year period (Conservation International 2023). These coefficients vary according to climatic region, ranging from 0.48 in tropical humid systems to 0.80 in temperate dry systems. Given the environmental characteristics of the SdEM, the tropical montane coefficient ($f = 0.64$) was applied in this study.

Relative SOC change was estimated by comparing baseline and target-year carbon stocks for each pixel according to the following equation:

$$SOC\ change\ (\%) = \frac{SOC_{2020} - SOC_{2001}}{SOC_{2001}} \times 100$$

Based on this relative variation, pixels were classified into three categories according to SDG 15.3.1 thresholds: areas were considered degraded when SOC decreased by 10% or more, improving when SOC increased by 10% or more, and stable when SOC variation ranged between − 10 and + 10%.

These thresholds are defined by the United Nations Convention to Combat Desertification (UNCCD) to ensure consistency in global land degradation assessments and to minimize the influence of short-term variability or data uncertainty on classification results (United Nations 2020; Conservation International 2023).

Integration of the SDG 15.3.1 indicator

The final SDG 15.3.1 indicator was obtained by integrating the three sub-indicators using the "one-out, all-out" principle, whereby a degradation signal in any sub-indicator results in classification of the area as degraded (Hu et al. 2021; Conservation International 2023) (Fig. 2).

Spatial analysis and modeling

Spatial autocorrelation of SDG 15.3.1 classes (degraded, stable, improving) was first evaluated using Global Moran's I to test for spatial dependence (Moran 1950; Tu and Xia 2008). Local Indicators of Spatial Association (LISA) were subsequently applied to identify spatial clusters and outliers (Anselin 1995; Harries 2006). Spatial weights were defined based on distance criteria, and cluster maps were generated to visualize spatial patterns.

Multinomial logistic regression (MLR) was used to model the relationship between environmental variables and SDG 15.3.1 classes (Maddala 1983; Long and Freese 2006). The stable class was used as the reference category. Independent variables included slope, roughness, hydrology, elevation, topographic

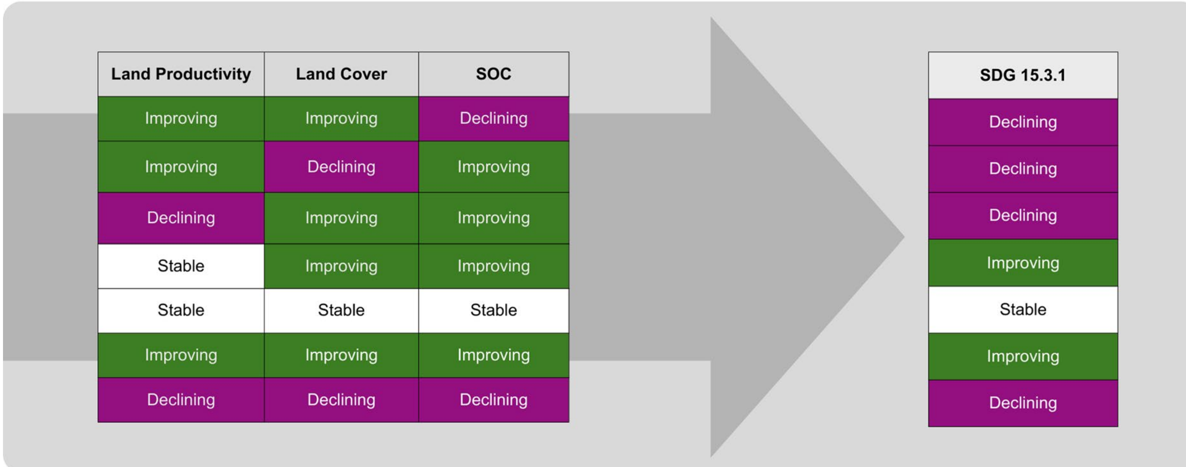

Fig. 2 Generation of the SDG 15.3.1 indicator based on the integration of sub-indicators (primary productivity, land cover, and soil organic carbon). The Y-axis represents the classification of land condition (degraded, stable, improving)

position index (TPI), and a spatial lag term derived from LISA results to account for residual spatial dependence (Anselin 1988; LeSage and Pace 2009).

Model fitting was performed using the multinom() function in the nnet package in R (R Core Team 2024; Venables and Ripley 2002). Model performance was evaluated using McFadden Pseudo-R^2 (McFadden 1972) and a multiclass confusion matrix. Overall accuracy, class-specific accuracy, and the Kappa coefficient were calculated to assess predictive performance (Fielding and Bell 1997).

Results and discussion

Primary productivity

Primary productivity patterns in the peatlands of the Serra do Espinhaço Meridional (SdEM) between 2001 and 2020 reveal a predominantly stable to improving vegetation dynamic, as evidenced by NDVI-based sub-indicators of productivity trajectory, state, and performance. These results indicate that, at the regional scale, peatland vegetation has largely maintained its functional capacity over the last two decades.

The productivity trajectory analysis shows that 57.27% (143.79 km^2) of the peatland area exhibited improving trends, while 40.77% (102.40 km^2) remained stable. Degrading trajectories were spatially restricted, affecting only 1.96% (4.92 km^2) of the total area. This predominance of positive and stable trajectories suggests long-term maintenance or recovery of vegetation cover, a key characteristic of peatlands where primary productivity is strongly regulated by hydrological conditions and organic matter accumulation.

Productivity state results further reinforce this pattern. Most peatlands (91.86%; 230.70 km^2) were classified as improving relative to historical productivity conditions, whereas only 1.11% (2.77 km^2) showed signs of degradation. This indicates that current productivity levels are generally higher than or comparable to long-term reference conditions, reflecting a high degree of ecological resilience. Such resilience is consistent with peatland systems that experience limited soil disturbance and sustained water saturation, which buffer vegetation productivity against short-term climatic variability.

Productivity performance exhibited the most homogeneous pattern, with 99.91% (250.91 km^2) of the peatland area showing no signs of degradation. This suggests that essential ecological processes related to biomass production remain functional across the SdEM peatlands, supporting peat accumulation and the long-term stability of these wetlands.

The integrated primary productivity indicator (Fig. 3; Table 2) confirms this overall pattern, with 50.65% (127.19 km^2) of the peatlands classified as improving and 47.55% (119.41 km^2) as stable.

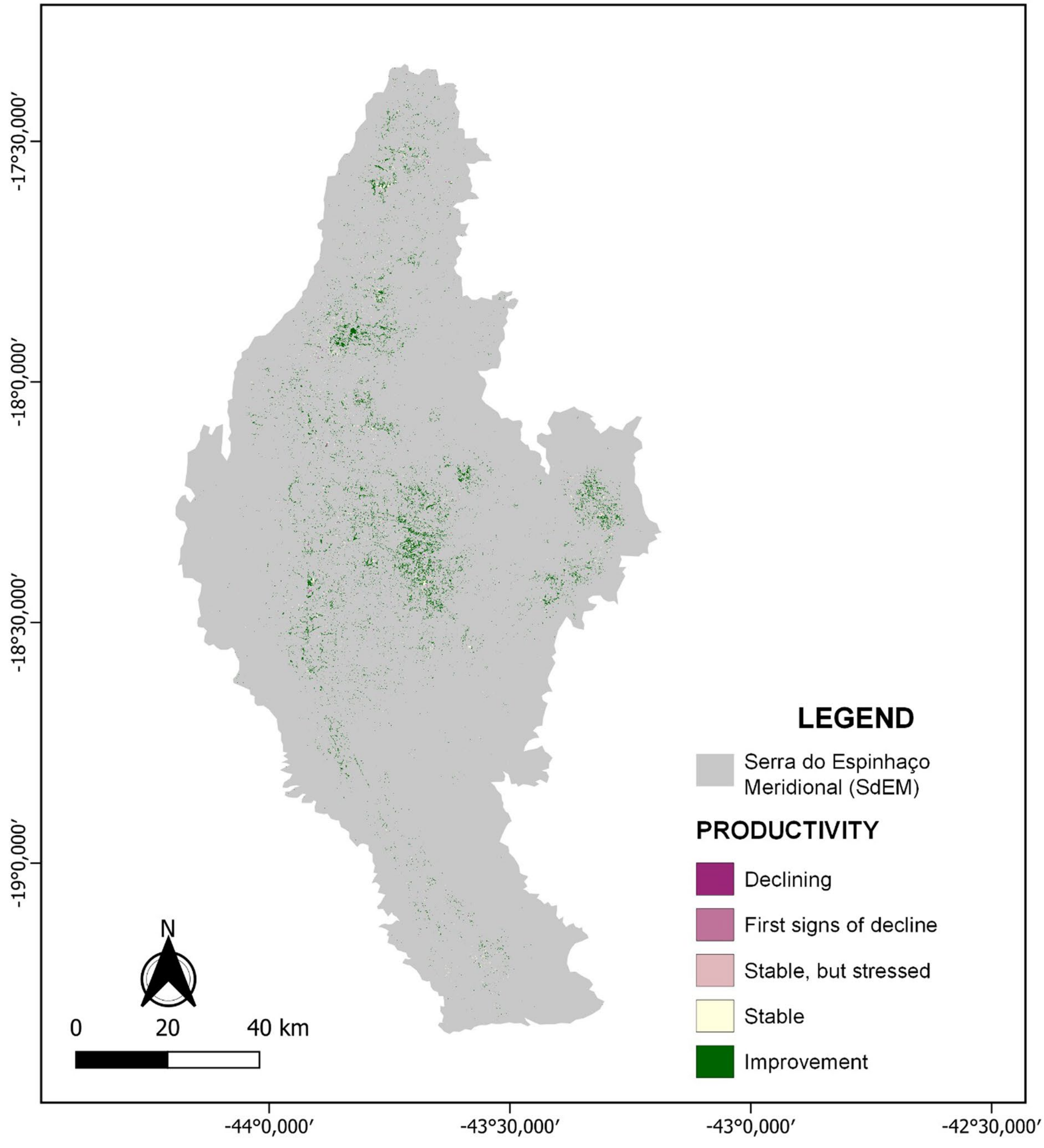

Fig. 3 Spatial distribution of peatland primary productivity from 2001 to 2020 in the Serra do Espinhaço Meridional, Minas Gerais, Brazil

Areas showing declining productivity were limited (1.48%), and only 0.31% exhibited early signs of decline. Together, these results indicate that SdEM peatlands have largely preserved their productive capacity, highlighting their role as resilient ecosystems under regional environmental change pressures.

Comparable findings have been reported for other tropical wetland systems. In the Pantanal, NDVI-based assessments revealed increasing productivity

Table 2 Area (km^2) of peatland primary productivity from 2001 to 2020 based on the calculation of the trajectory, state, and performance sub-indicators in the Serra do Espinhaço Meridional, Minas Gerais, Brazil

Condition	Area (Km^2)	Percentage (%)
Declining	3,72	1,48
First signs of decline	0,76	0,31
Stable, but stressed	3,72	1,48
Stable	119,41	47,55
Improving	127,19	50,65
Total	251,13	100

trends linked to climatic drivers such as temperature and precipitation (Demarquet et al. 2024). Similarly, long-term studies in Chinese wetlands demonstrated sustained productivity and gradual vegetation recovery under environmental stress (Liu et al. 2015). These parallels support the robustness of NDVI-derived indicators for assessing productivity dynamics in peatland ecosystems.

Land cover

Land cover dynamics in the peatlands of the Serra do Espinhaço Meridional (SdEM) between 2001 and 2020 indicate a high degree of structural stability (Fig. 4; Table 3). The vast majority of the peatland area (98.95%; 248.47 km^2) remained stable over the study period, highlighting the persistence of land cover patterns and the limited extent of anthropogenic disturbance within these wetland systems.

Land cover degradation was extremely restricted, affecting only 0.09% (0.21 km^2) of the total peatland area. In contrast, improving land cover conditions were identified in 0.97% (2.43 km^2), suggesting localized recovery processes. These improving areas are likely associated with natural regeneration dynamics, such as the expansion or densification of woody and herbaceous vegetation under favorable hydrological conditions.

The predominance of stable land cover reflects the inherent resilience of high-altitude tropical peatlands, where persistent soil saturation, low accessibility, and environmental protection constraints limit land-use conversion. Similar stability patterns have been reported in other peatland and wetland systems, where minimal land cover change is indicative of intact ecological structure and sustained ecosystem functioning (Parish et al. 2008; Silva et al. 2022).

Overall, the land cover condition results corroborate the primary productivity findings, reinforcing the conclusion that SdEM peatlands have maintained their ecological integrity over the last two decades. This stability is particularly relevant for peatland conservation, as land cover integrity is directly linked to the preservation of hydrological regulation, carbon storage, and biodiversity in wetland ecosystems.

Soil organic carbon

Soil organic carbon (SOC) dynamics in the peatlands of the Serra do Espinhaço Meridional (SdEM) between 2001 and 2020 indicate an overwhelmingly stable carbon system (Fig. 5; Table 4). Approximately 99.95% of the peatland area remained classified as stable, demonstrating that carbon stocks were effectively maintained over the two decades analyzed. This stability reflects the persistence of environmental conditions that favor peat accumulation, such as long-term soil saturation, limited disturbance, and continuous organic matter inputs.

SOC degradation was detected in only 0.02% of the peatland area, while improving conditions accounted for 0.03%. Although spatially limited, these deviations from stability are ecologically relevant. Localized SOC degradation may be associated with site-specific disturbances, including drainage, fire, trampling, or hydrological alteration, which are known to accelerate peat oxidation and carbon losses. Conversely, improving SOC conditions likely reflect enhanced organic matter accumulation driven by favorable hydrological regimes or natural recovery processes.

The predominance of stable SOC highlights the role of SdEM peatlands as effective long-term carbon sinks and reinforces their importance for regional climate regulation. Similar patterns have been documented in tropical peatlands under minimal anthropogenic pressure, where SOC stocks remain largely stable over time (Suhaili et al. 2021). In contrast, studies from Southeast Asia demonstrate that peatlands subjected to drainage and land-use conversion experience substantial carbon losses, underscoring the vulnerability of SOC stocks to human disturbance (Murdiyarso et al. 2010).

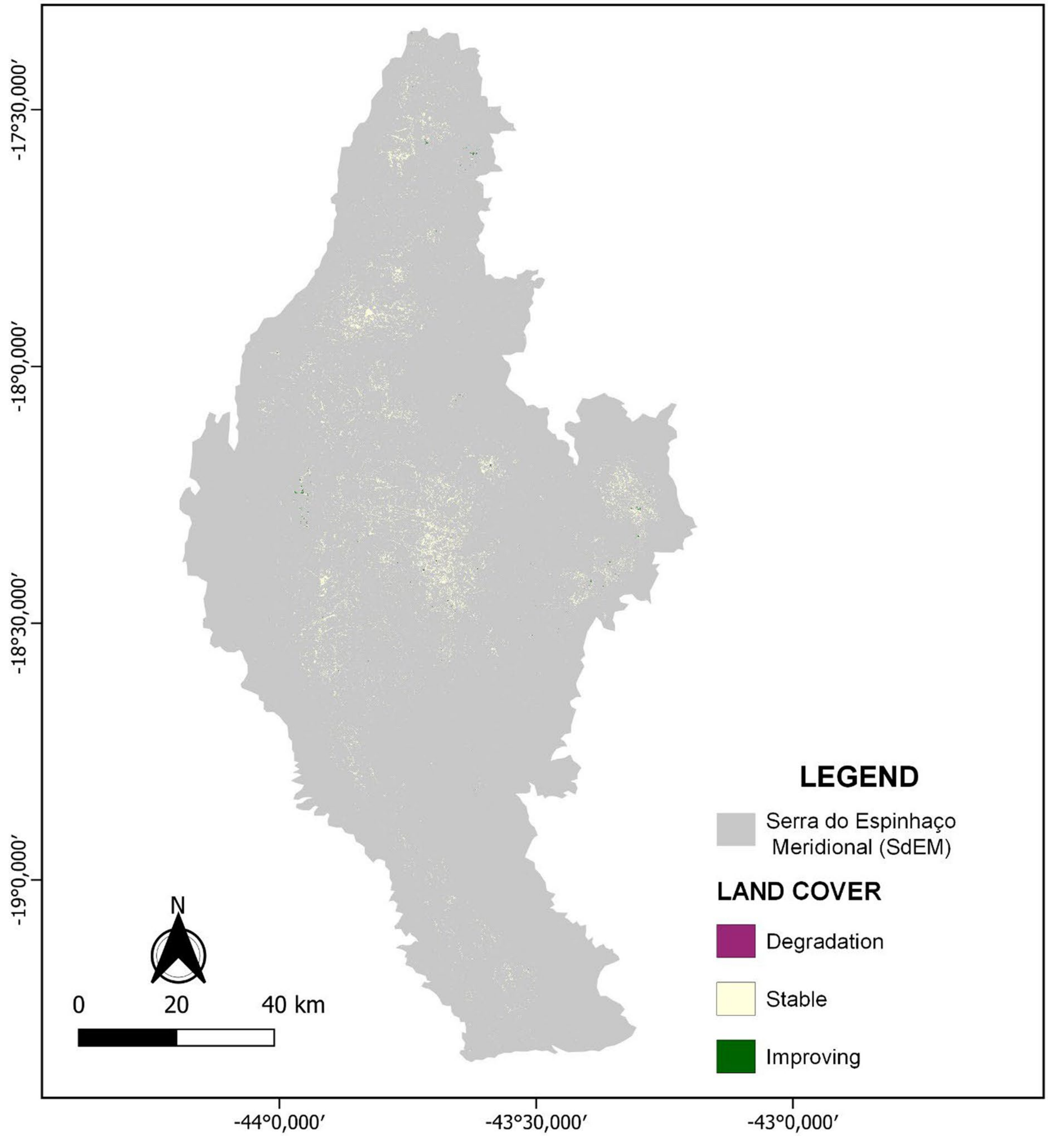

Fig. 4 Land cover condition of peatlands in the Serra do Espinhaço Meridional, Minas Gerais, from 2001 to 2020

The results also demonstrate the applicability of integrated remote sensing-based approaches for detecting spatial and temporal variations in peatland carbon dynamics. The ability to identify even small fractions of SOC degradation or improvement is particularly relevant for monitoring and conservation planning, as early detection of carbon loss can inform preventive management actions. Overall, the SOC assessment confirms the high ecological integrity of SdEM peatlands while highlighting the need for continued protection of their hydrological functioning to safeguard long-term carbon storage.

Table 3 Land cover condition of peatlands in the Serra do Espinhaço Meridional, Minas Gerais, from 2001 to 2020

Condition	Area (Km^2)	Percentage (%)
Stable	248,47	98,95
Degradation	0,21	0,09
Improving	2,43	0,97
Total	251,13	100

Integrated SDG 15.3.1 assessment and spatial patterns

The integration of primary productivity, land cover, and soil organic carbon through the SDG Indicator 15.3.1 revealed a predominantly positive conservation status for the peatlands of the Serra do Espinhaço Meridional (Fig. 6). More than half of the assessed

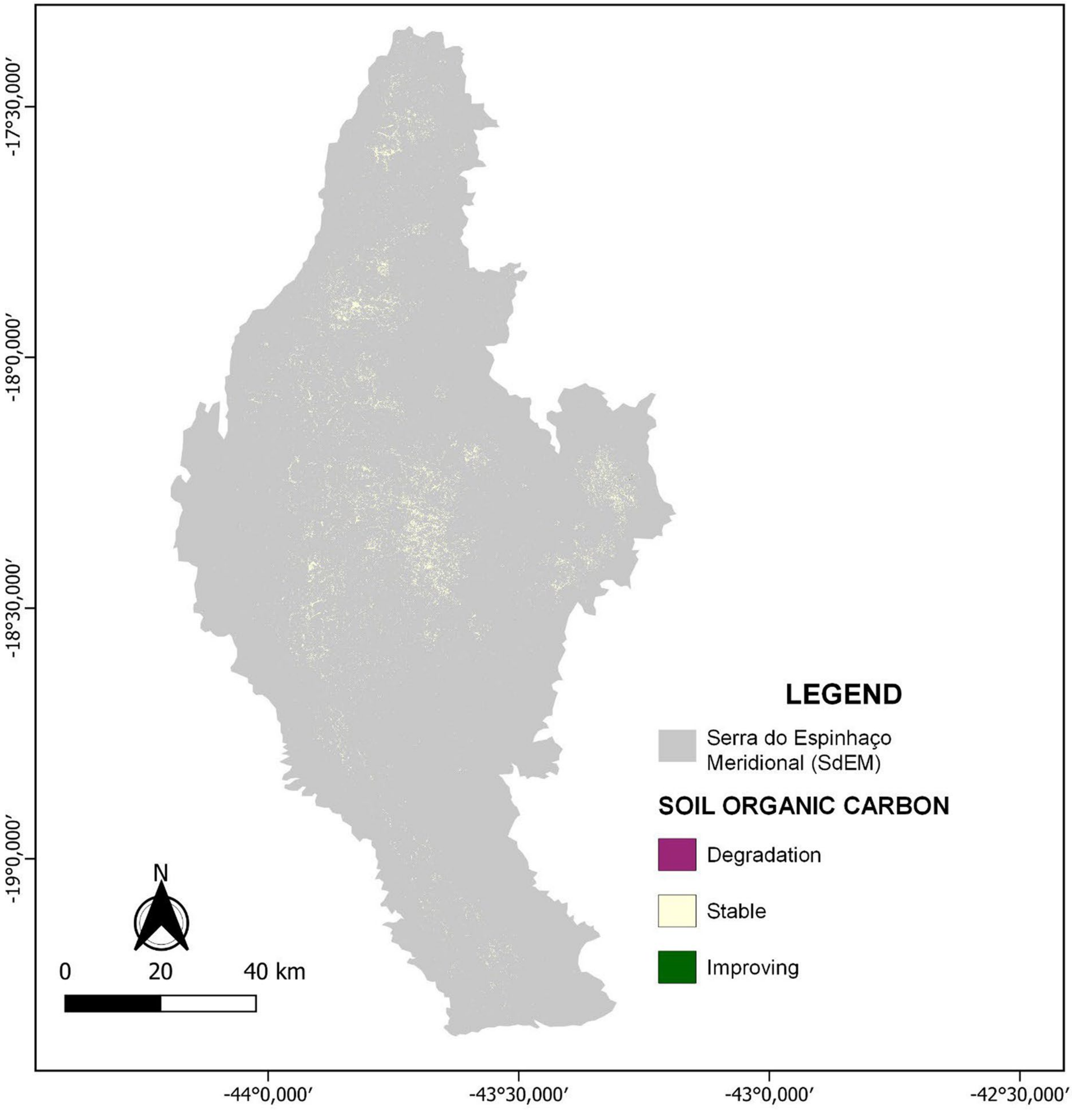

Fig. 5 Soil organic carbon of peatlands in the Serra do Espinhaço Meridional, Minas Gerais, for 2001 and 2020

Table 4 Variation in soil organic carbon in peatlands of the Serra do Espinhaço Meridional, Minas Gerais, from 2001 to 2020

Condição	Area (Km^2)	Percentagem (%)
Stable	251,01	99,95
Degradation	0,04	0,02
Improving	0,06	0,03
Total	251,13	100

area was classified as improving, while nearly half remained stable. Degradation affected only a small fraction of the peatland area (Table 5).

This integrated assessment highlights the coherence among sub-indicators, particularly the strong influence of primary productivity and SOC stability on the final indicator. The dominance of stable and improving classes reflects favorable ecological conditions, low land-use intensity, and the persistence of hydrological regimes essential for peatland functioning. Similar applications of SDG 15.3.1 in other regions have shown that wetlands and areas with limited anthropogenic disturbance tend to exhibit positive or stable degradation trends (Hu et al. 2021; Jiang et al. 2022).

The results emphasize the usefulness of SDG 15.3.1 as a diagnostic tool for wetlands, enabling the identification of conservation priorities and areas requiring monitoring. In the SdEM, the indicator confirms that peatlands remain largely functional ecosystems, supporting carbon storage, water regulation, and biodiversity maintenance.

Environmental drivers of degradation and recovery (spatial modeling)

Spatial analysis revealed a strong and statistically significant spatial structure in the distribution of SDG 15.3.1 classes. Global and local Moran's I analyses demonstrated pronounced clustering of similar conditions, with extensive High–High clusters corresponding to areas classified as improving (Fig. 7). These clusters were predominantly located in the northern and northeastern portions of the study area, indicating large, contiguous peatland systems with high environmental integrity (Table 6).

In contrast, degradation-related clusters were sparse and spatially isolated, suggesting localized disturbances rather than widespread degradation processes. Transitional Low–High clusters occurred infrequently and may represent ecotonal or sensitive zones where small changes in hydrology or land use could trigger shifts in ecosystem condition. Such spatial configurations are characteristic of peatlands, where ecological responses are tightly coupled to topography and water availability (Page et al. 2011).

Multinomial logistic regression results further demonstrated that topographic and hydrological variables play a key role in shaping peatland condition. Slope, roughness, elevation, hydrology, and topographic position index were all significant predictors of SDG 15.3.1 classes (Table 7). Areas with low slope and smoother terrain were more susceptible to degradation, likely due to greater accessibility and anthropogenic influence, whereas more rugged and elevated areas favored stability or improvement.

The inclusion of a spatial lag term substantially improved model performance, confirming that peatland condition is not spatially independent and that neighboring areas exert strong influence on local dynamics. This finding aligns with spatial ecology theory and highlights the importance of explicitly accounting for spatial dependence when modeling wetland degradation and recovery processes (Anselin 1988; LeSage and Pace 2009).

Model performance metrics indicated high predictive accuracy for stable and improving classes, while degradation was less accurately predicted due to its low prevalence. This limitation is consistent with challenges commonly reported in modeling rare environmental events (King and Zeng 2001). Nonetheless, the overall results demonstrate that spatial modeling provides robust insights into the drivers and patterns of peatland condition across the SdEM.

Uncertainties and methodological limitations

This study is subject to uncertainties related to the spatial resolution and intrinsic limitations of the geospatial datasets used. The MODIS NDVI (250 m) and ESA CCI land cover (300 m) products are widely applied in regional environmental assessments; however, their moderate spatial resolution may not fully capture the fine-scale heterogeneity typical of tropical montane peatlands. In these environments, individual pixels often represent mixed land-cover signals, which may smooth local variability and reduce

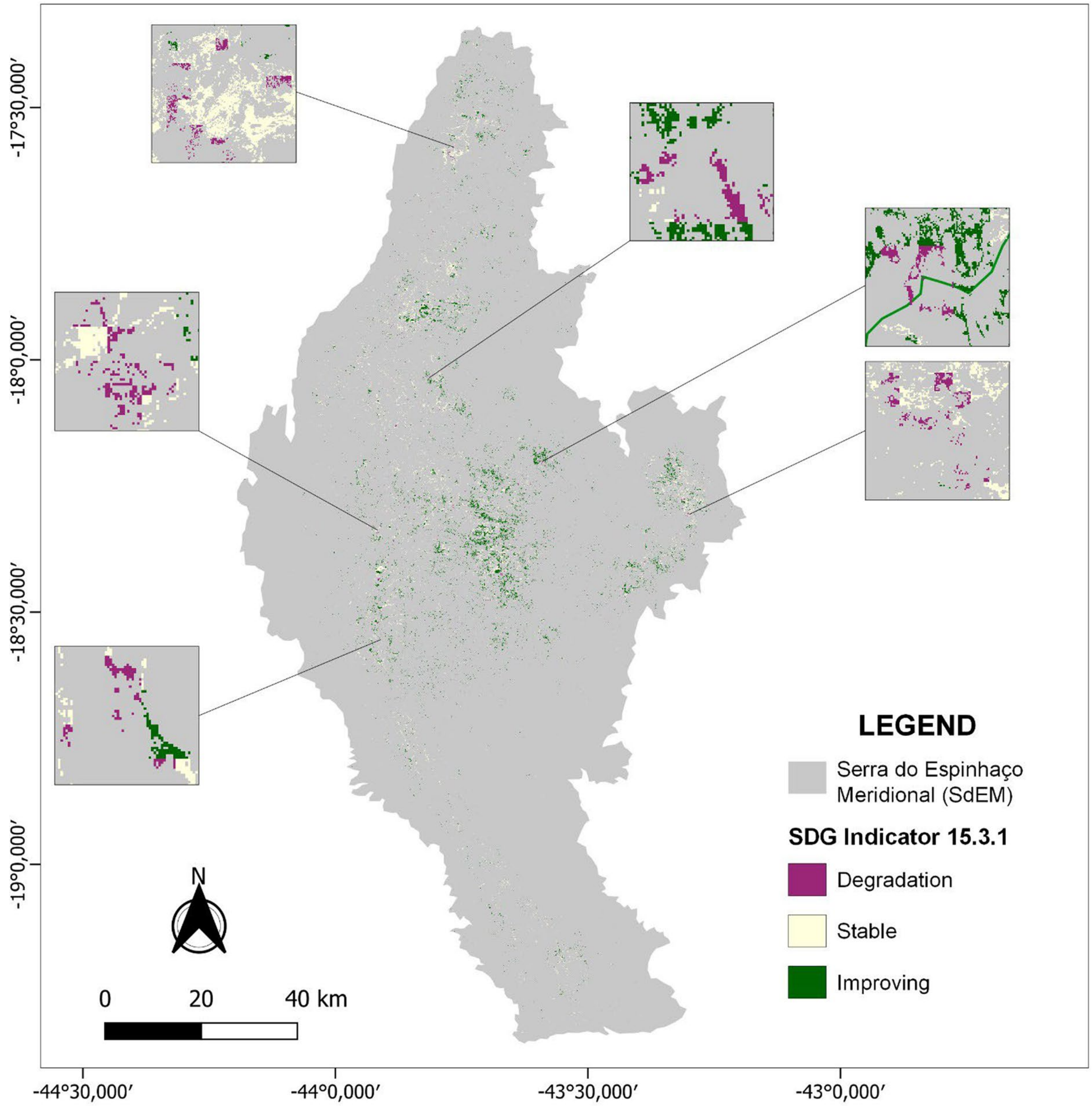

Fig. 6 Results of SDG Indicator 15.3.1 for peatlands in the Serra do Espinhaço Meridional, Minas Gerais, for 2001 and 2020

the detection of small or fragmented peatland patches (Hu et al. 2021; Giuliani et al. 2020).

At this scale, pixel values represent average conditions within each cell. Therefore, localized degradation processes or small peatland areas are not entirely absent from the analysis but may be diluted within the aggregated signal, reducing sensitivity to fine-scale spatial patterns. This limitation is particularly relevant in heterogeneous landscapes, where NDVI-based indicators are known to be affected by scale-related biases (Ivits and Cherlet 2013; Easdale et al. 2019). Likewise, MODIS-derived products may have limited capacity to resolve sub-pixel land-cover composition, whereas higher-resolution sensors such as Sentinel-2 are more suitable for detailed ecological assessments (Reith et al. 2021).

Table 5 SDG Indicator 15.3.1 obtained from the integration of primary productivity, land cover, and soil organic carbon for the Serra do Espinhaço Meridional, Minas Gerais, Brazil

Condition	Area (Km^2)	Percentage (%)
Stable	118,40	47,15
Degradation	4,67	1,86
Improving	128,03	50,98
Total	251,13	100

Additional uncertainties are associated with SoilGrids, which is based on machine learning models trained on globally distributed soil profile datasets and environmental covariates (Poggio et al. 2021). The limited representation of tropical montane environments in the training database, together with the 250 m resolution, may underestimate soil organic carbon variability in small peatland systems and lead to conservative estimates of degradation or improvement (Poggio et al. 2021; Turek et al. 2023). Furthermore, within the SDG 15.3.1 framework, SOC dynamics are indirectly estimated through the integration of baseline SOC stocks with land-cover transitions and standardized conversion factors rather than direct repeated soil measurements (Conservation International 2023). Although this combined approach enables globally consistent assessments, it may oversimplify local SOC responses by assuming generalized relationships between land-use change and carbon dynamics, potentially overlooking site-specific ecological processes, management practices, or short-term fluctuations. Consequently, SOC estimates should be interpreted as approximations of broader degradation trends rather than precise measurements of local carbon stock variation.

Future vulnerability under climate and land-use change

Although the results indicate a predominantly stable and improving condition of peatlands in the Serra do Espinhaço Meridional (SdEM), their long-term persistence may be challenged by projected climate and land-use changes in southeastern Brazil. Regional projections suggest increasing temperatures, shifts in precipitation seasonality, and more frequent drought events, which may significantly affect peatland hydrology (Intergovernmental Panel on Climate Change IPCC 2023; Marengo et al. 2017). As peatland functioning depends on sustained water saturation, reductions in water availability can lower water tables, enhance aerobic decomposition, and accelerate peat oxidation, leading to carbon loss and ecosystem degradation (Dise 2009; Murdiyarso et al. 2010; Page et al. 2011).

In the SdEM, peatland distribution is strongly controlled by geomorphological and hydrological conditions, particularly the interaction between quartzitic substrates, drainage patterns, and water retention capacity (Silva et al. 2009; Rocha Campos et al. 2016). While these controls support long-term organic matter accumulation, they also increase sensitivity to environmental changes, as even small alterations in drainage conditions can affect peat functioning (Silva et al. 2009; Holden et al. 2004). This vulnerability is reinforced by the high water storage capacity of these systems, which depend on hydrological stability (Silva et al. 2013; Joosten and Clarke 2002; Dise 2009).

Land-use change represents an additional risk. Although direct anthropogenic pressure in the SdEM remains relatively low, the expansion of agriculture, infrastructure, and fire occurrence may alter drainage and vegetation dynamics. Even localized disturbances can propagate through hydrological networks and affect peatland stability (Rocha Campos et al. 2016), and evidence from tropical peatlands shows that minor hydrological changes can result in substantial carbon losses (Murdiyarso et al. 2010; Page et al. 2011). Fire is an additional concern, as drought-related peat fires can release large amounts of stored carbon and cause long-term ecological damage (Page et al. 2002; Turetsky et al. 2015).

Conclusions

This study provides an integrated and spatially explicit assessment of land degradation dynamics in tropical montane peatlands of the Serra do Espinhaço Meridional using the SDG Indicator 15.3.1 combined with remote sensing, spatial statistics, and environmental modeling. At the regional scale, peatlands

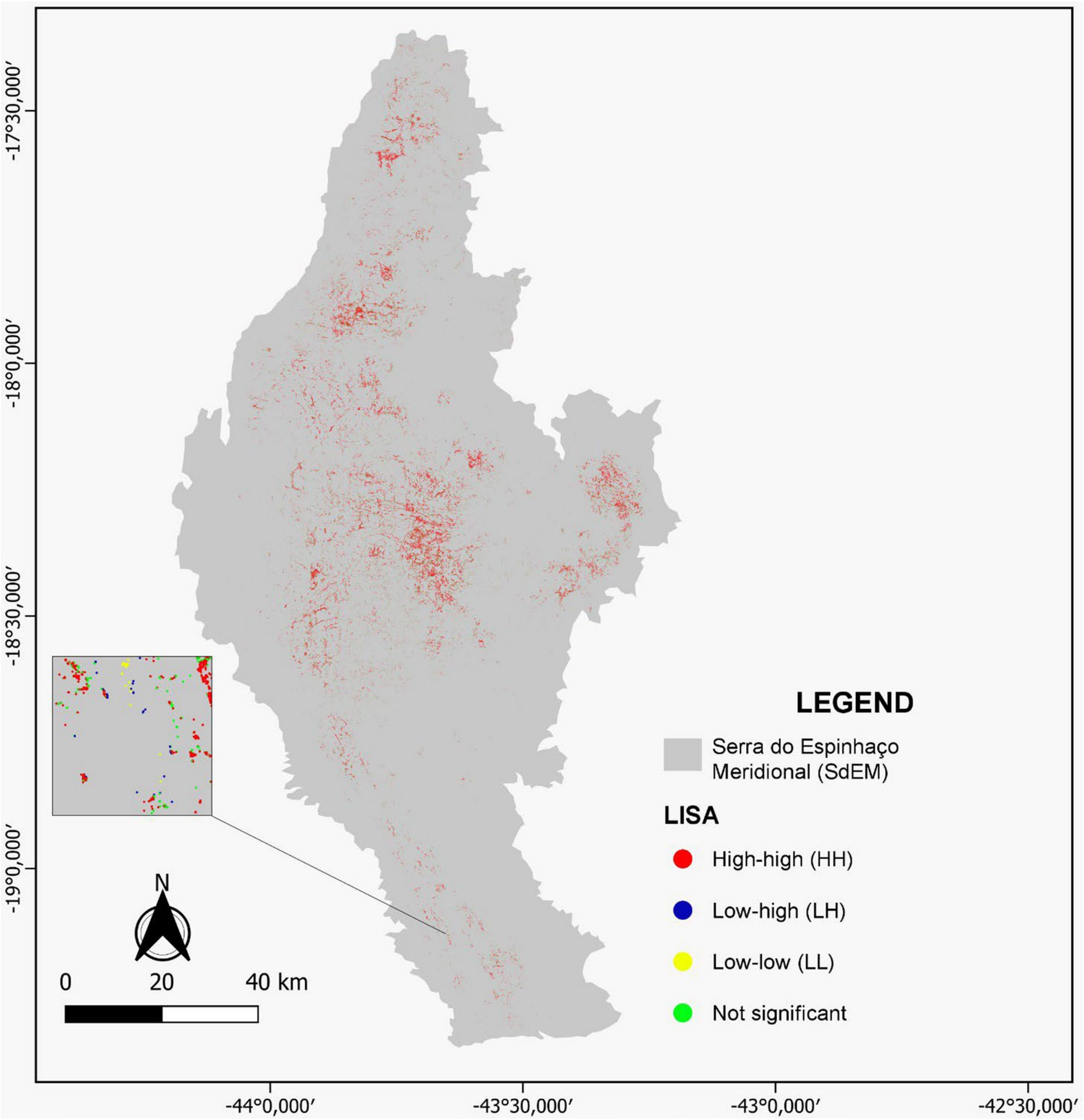

Fig. 7 Spatial distribution of local spatial autocorrelation clusters (LISA) of the SDG 15.3.1 indicator in the peatlands of the Serra do Espinhaço Meridional

were predominantly classified as stable or improving, while degradation was limited and spatially clustered, indicating localized rather than widespread ecosystem impairment.

The combined analysis of primary productivity, land cover, and soil organic carbon revealed a high degree of ecological resilience. Vegetation productivity showed predominantly stable and improving trends, land cover remained largely unchanged over

Table 6 Distribution of LISA cluster analysis in the peatlands of the Serra do Espinhaço Meridional

Clusters	Numbers of features	Km^2
High-high (HH)	558.983 (84,1%)	211,07 Km^2
Low-high (LH)	47 (0,007%)	0,02 Km^2
Low-Low (LL)	33 (0,005%)	0,01 Km^2
Not significant	105.548 (15,9%)	39,99 Km^2
Total		251,13 Km^2

two decades, and soil organic carbon stocks were maintained across nearly the entire peatland area. These results confirm the capacity of these ecosystems to sustain key functions, particularly long-term carbon storage and hydrological regulation, under current land-use conditions.

Spatial analyses demonstrated that peatland condition is strongly structured by spatial dependence, with hydrological and topographic variables emerging as primary drivers of degradation, stability, and improvement. Areas with greater hydrological stability and terrain heterogeneity exhibited higher resilience, whereas more accessible and low-relief areas showed increased susceptibility to degradation.

The application of the SDG 15.3.1 indicator proved to be a consistent and transferable approach for diagnosing peatland condition and identifying priority areas for monitoring and conservation. However, the results should be interpreted considering the limitations associated with moderate-resolution geospatial datasets and the uncertainties inherent to global products, which may reduce sensitivity to fine-scale spatial variability.

Although current conditions indicate resilience, the long-term stability of these peatlands may be challenged by projected climate change and increasing land-use pressures. Changes in hydrological regimes, drought frequency, and disturbance processes may enhance vulnerability and affect ecosystem functioning, reinforcing the need for proactive conservation strategies.

Overall, this study fills an important knowledge gap regarding tropical montane peatlands and provides a scalable framework to support monitoring, conservation planning, and sustainable land management. Maintaining hydrological integrity emerges as a central priority to ensure the persistence of ecosystem services and the long-term stability of these systems under future environmental change.

Table 7 Comparison of multinomial logistic regression results with and without the *lag_value* term for SDG 15.3.1

Metric/variable	Model without lag_value	Model with lag_value
Log-verossimilhança	−512.138,4	−160.592,7
Pseudo-R^2 of McFadden	—	0,6876
Slope	$\beta = -0{,}000220246$ $p < 0{,}001$	$\beta = -0{,}000221362$ $p < 0{,}001$
Roughness	$\beta = 35{,}77{,}454$ $p < 0{,}001$	$\beta = 35{,}36{,}575$ $p < 0{,}001$
Hidrology	$\beta = 0{,}01955100$ $p < 0{,}001$	$\beta = 0{,}01956380$ $p < 0{,}001$
Altitude	$\beta = 0{,}08663363$ $p < 0{,}001$	$\beta = 0{,}08095524$ $p < 0{,}001$
TPI	$\beta = -4{,}822{,}364$ $p < 0{,}001$	$\beta = -4{,}884{,}700$ $p < 0{,}001$
lag_value	—	$\beta = 7{,}03043919$ $p < 0{,}001$
Test LR (Chi^2)	—	3.703.091,4 ($p < 0{,}001$)

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Author contributions Maria Fernanda Ferreira Carvalho contributed to the study conception and design, data acquisition, spatial analysis, environmental modeling, interpretation of results, and writing of the original draft. Cristiano Christofaro contributed to the study conception, methodological design, interpretation of results, and critical revision of the manuscript. Alexandre Christofaro Silva contributed to data interpretation, discussion of results, and critical review of the manuscript. Uidemar Morais Barral contributed to methodological support, spatial analysis interpretation, and manuscript revision. All authors read and approved the final manuscript.

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Declarations

Competing interests The authors declare no competing interests.

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